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LearninGT
Will | Knowledge | Action

Generative AI for Business

Session: 1

#GTBharat
SHAPING VIBRANT INDIA

Artificial Intelligence

Understanding AI

What makes us humans ?

- Hearing
- Sight
- Touch
- Taste
- Smelling

And what makes us intelligent ?

- Learning
- Reasoning
- Problem Solving
- Social Behavior
- Creativity

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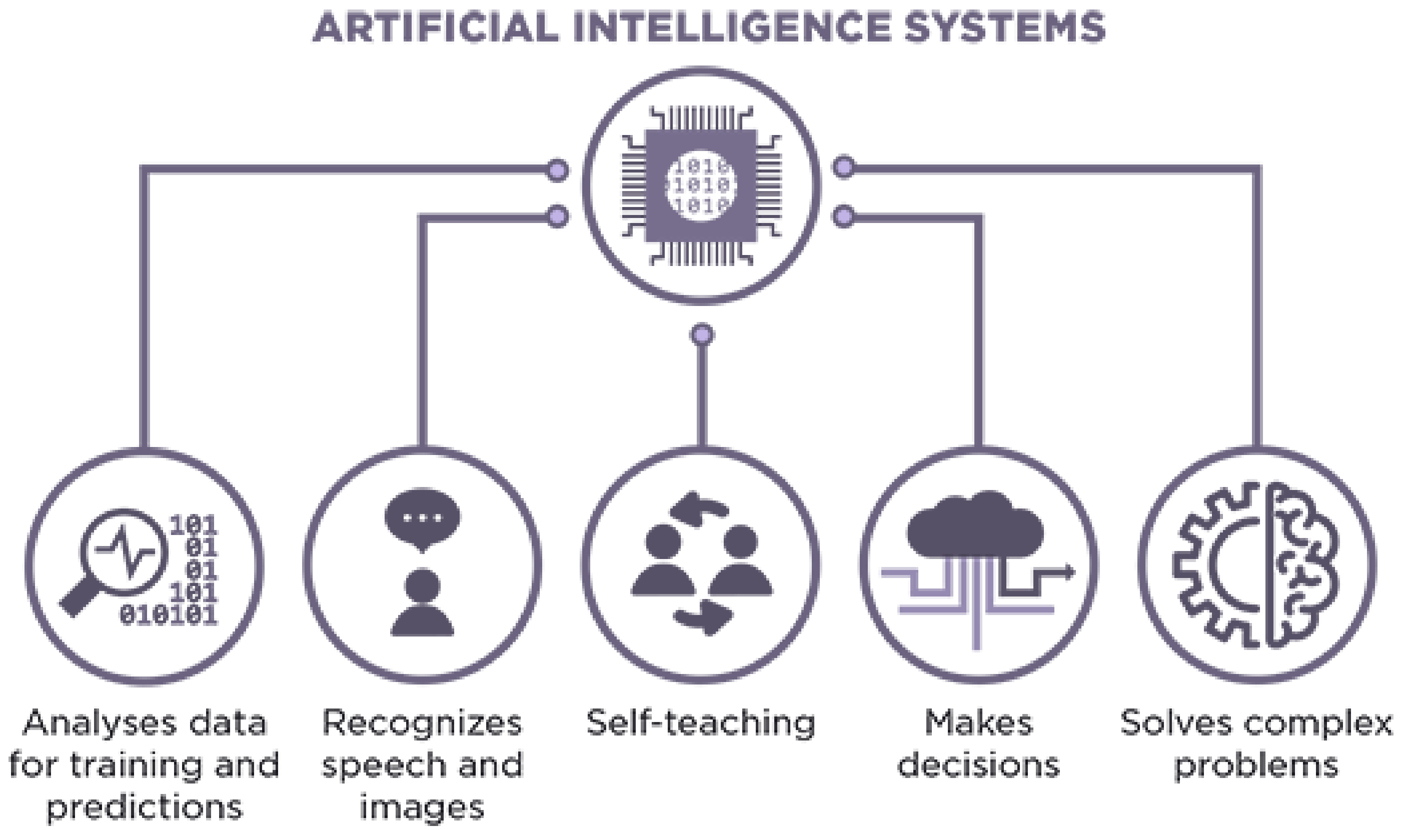
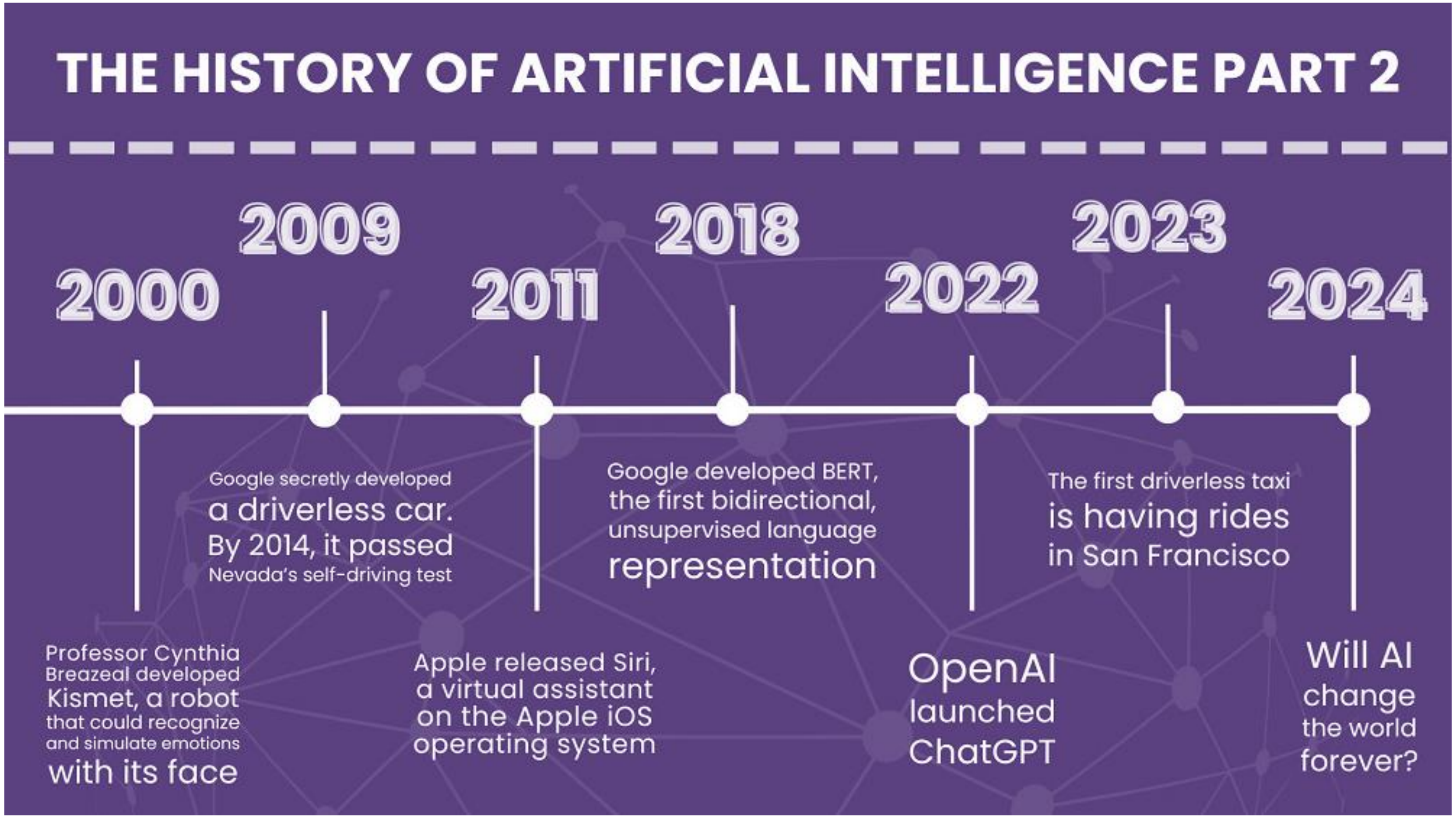
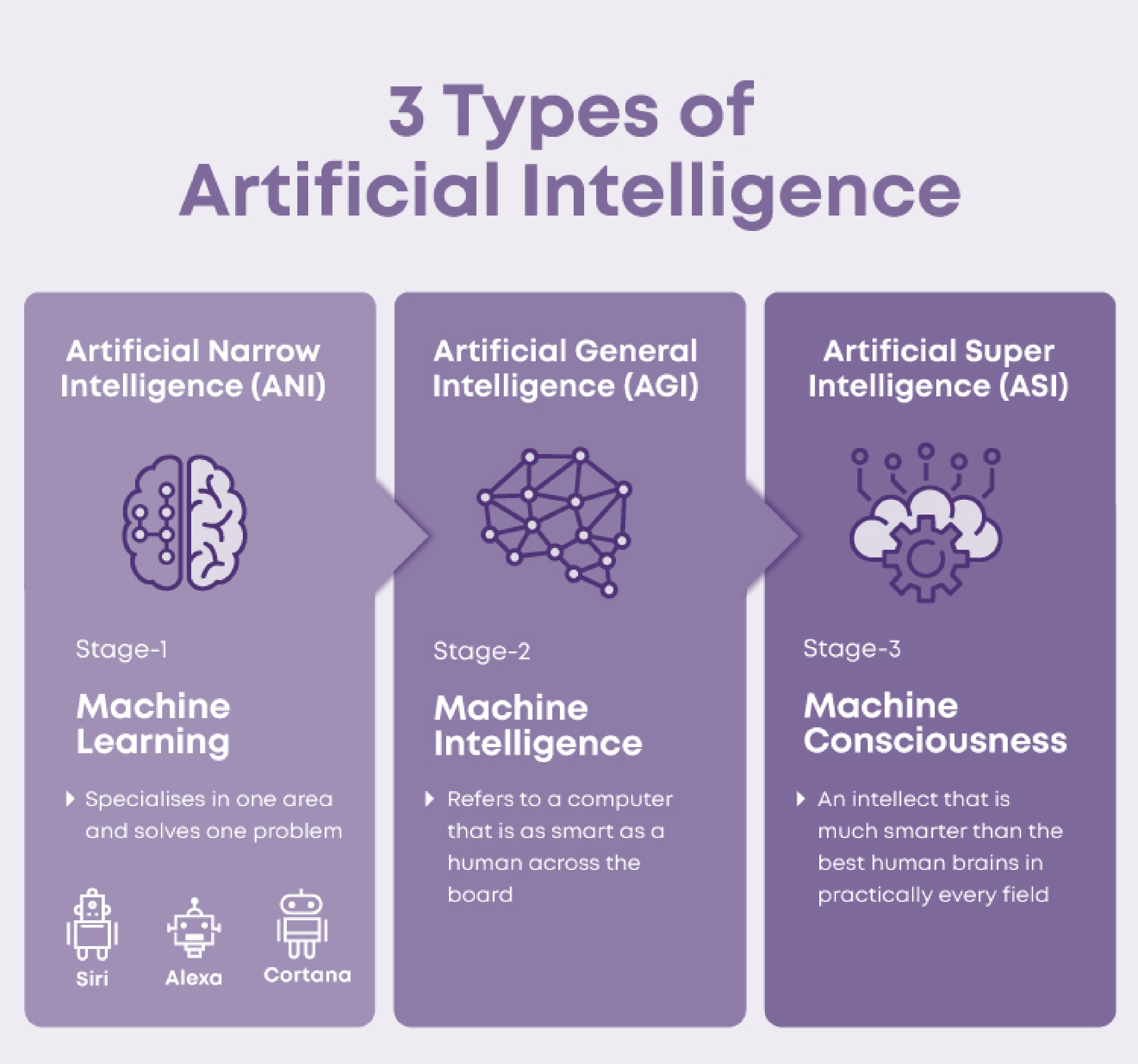


- **Artificial Intelligence** (AI), sometimes called machine intelligence, is intelligence demonstrated by machines, in contrast to the natural intelligence displayed by humans and other animals, such as "learning" and "problem solving"

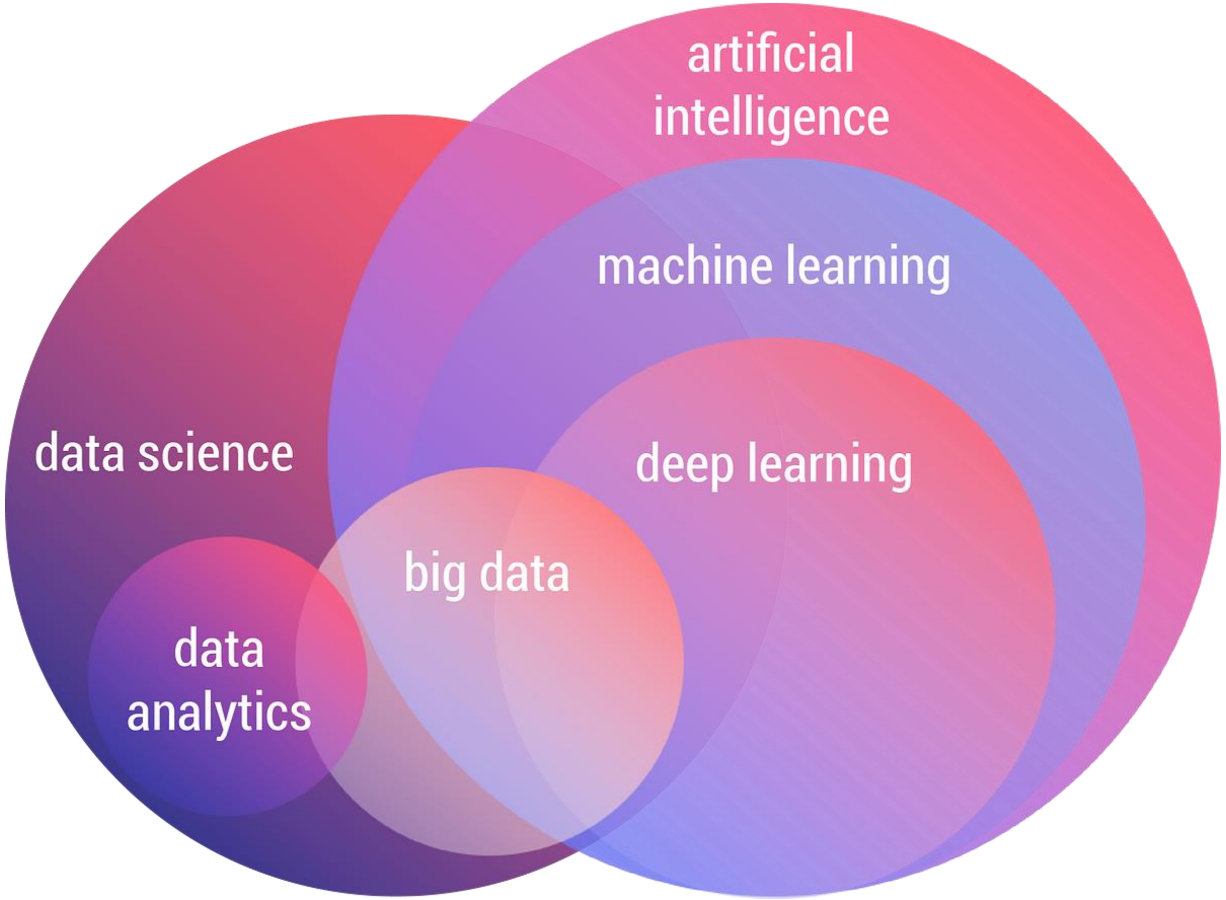
- In computer science, **AI research** is defined as the study of "intelligent agents": any device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals.

History of New Edge AI

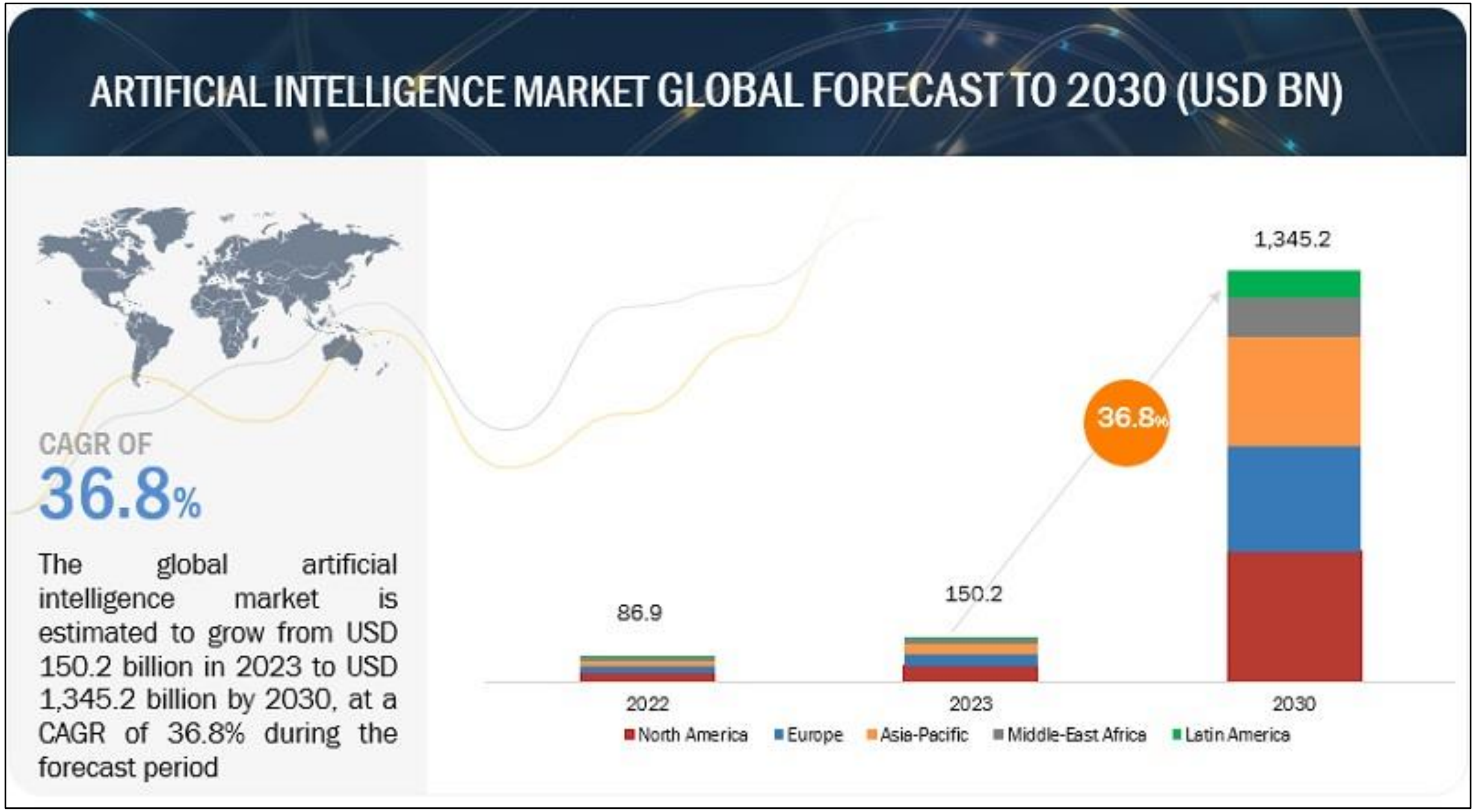



3 Types of
Artificial Intelligence
Artificial Narrow Intelligence (ANI)
Stage-1
Machine Learning
Specialises in one area and solves one problem
Siri
Alexa
Cortana
Artificial General Intelligence (AGI)
Stage-2
Machine Intelligence
Refers to a computer that is as smart as a human across the board
Artificial Super Intelligence (ASI)
Stage-3
Machine Consciousness
An intellect that is much smarter than the best human brains in practically every field

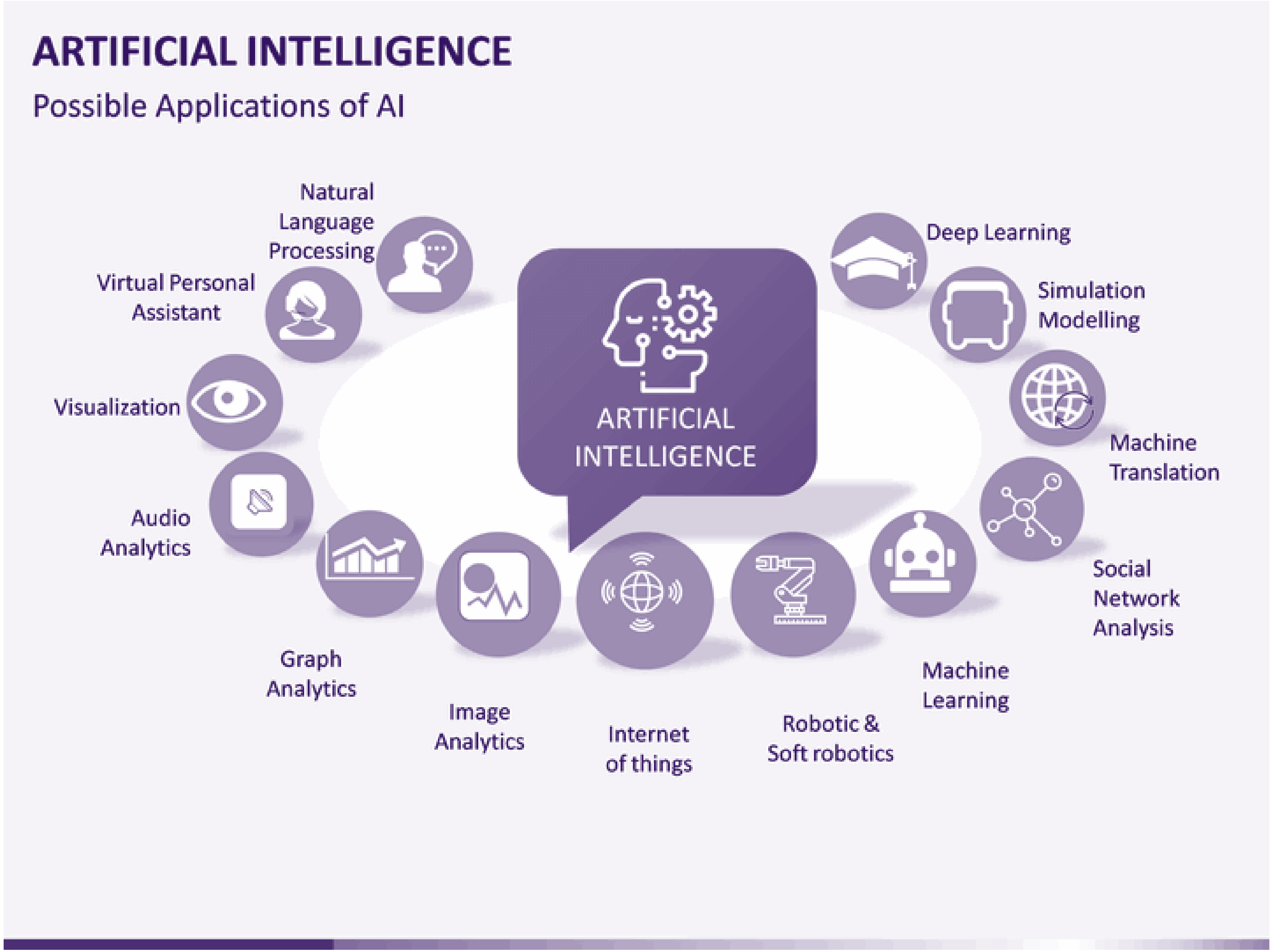
The Complete Domain



Growth of AI



ARTIFICIAL INTELLIGENCE MARKET GLOBAL FORECAST TO 2030 (USD BN)
CAGR OF
36.8%
The global artificial intelligence market is estimated to grow from USD 150.2 billion in 2023 to USD 1,345.2 billion by 2030, at a CAGR of 36.8% during the forecast period
86.9
150.2
1,345.2
36.8%
2022
2023
2030
North America
Europe
Asia-Pacific
Middle-East Africa
Latin America


ARTIFICIAL INTELLIGENCE
Possible Applications of AI
Natural Language Processing
Virtual Personal Assistant
Visualization
Audio Analytics
Graph Analytics
Image Analytics
Internet of things
Robotic & Soft robotics
Machine Learning
Social Network Analysis
Machine Translation
Simulation Modelling
Deep Learning
ARTIFICIAL INTELLIGENCE

Top Use Cases of AI across Industries

General	Healthcare	Retail	Financial Services	Professional Services
Resume Matching	Real Time Pregnancy Risk Evaluation	Packaging Quality Evaluation	Fraud Detection	RFP Opportunities Classification
Help Desk Answers	Patient Receivables Management	Inventory Management	Personal Loan Approval	Deal Guidance
Customer Churn Prediction	Propensity of Claim Denial Prediction	Merchandising Planning	KYC – Entity Identification	Data Extraction from Charts
Customer Complaints – Email Classification	Fraudulent Medical Claim Prediction	Product Recommendation	AML Alert Classification	Auditing – Anomaly Detection
Quality – Visual Inspection	Readmission Prediction	Pricing Optimization	ID Information Extraction	Legal – Win/Loss Rate Prediction

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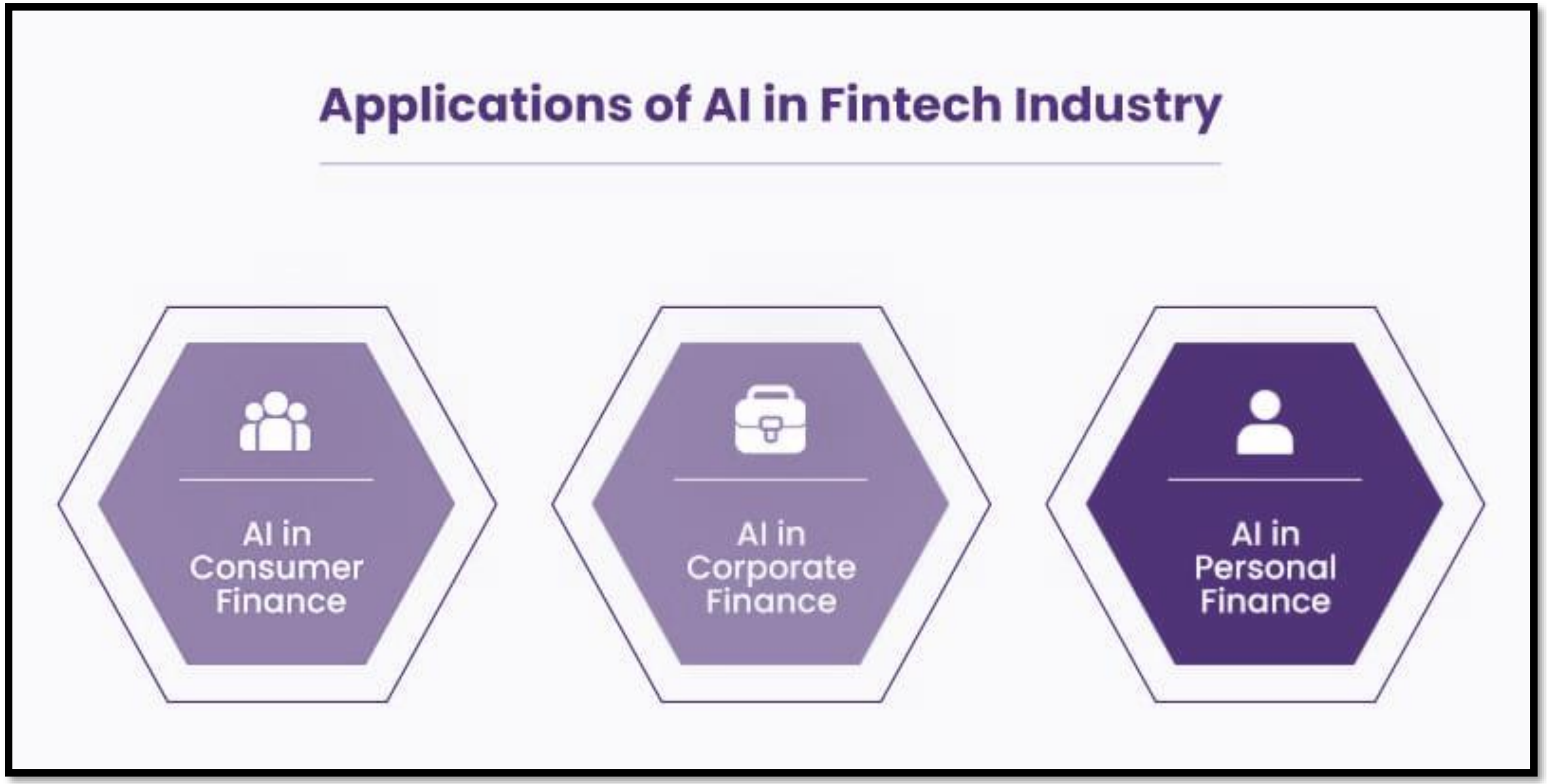
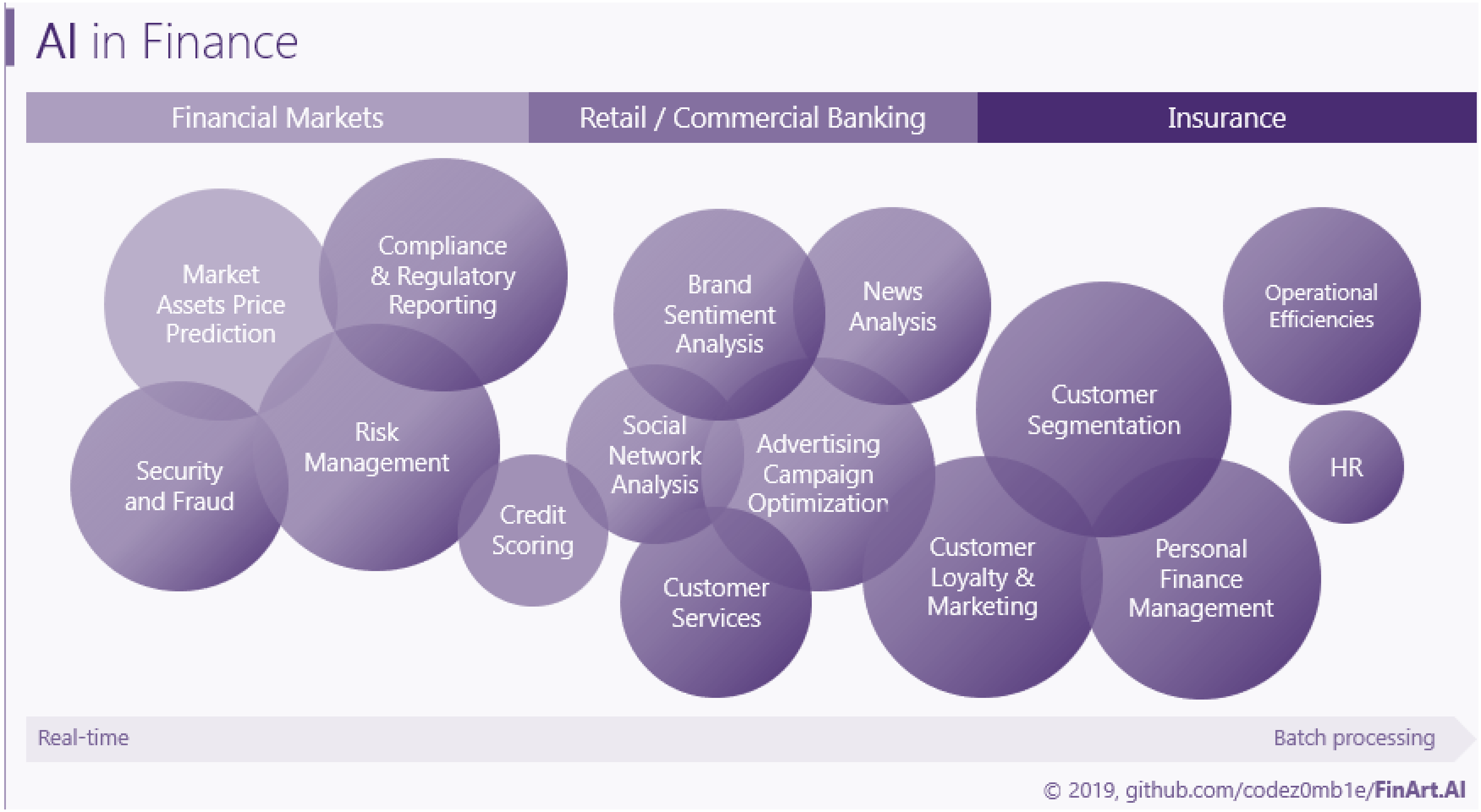
Where can you use AI in Finance ?

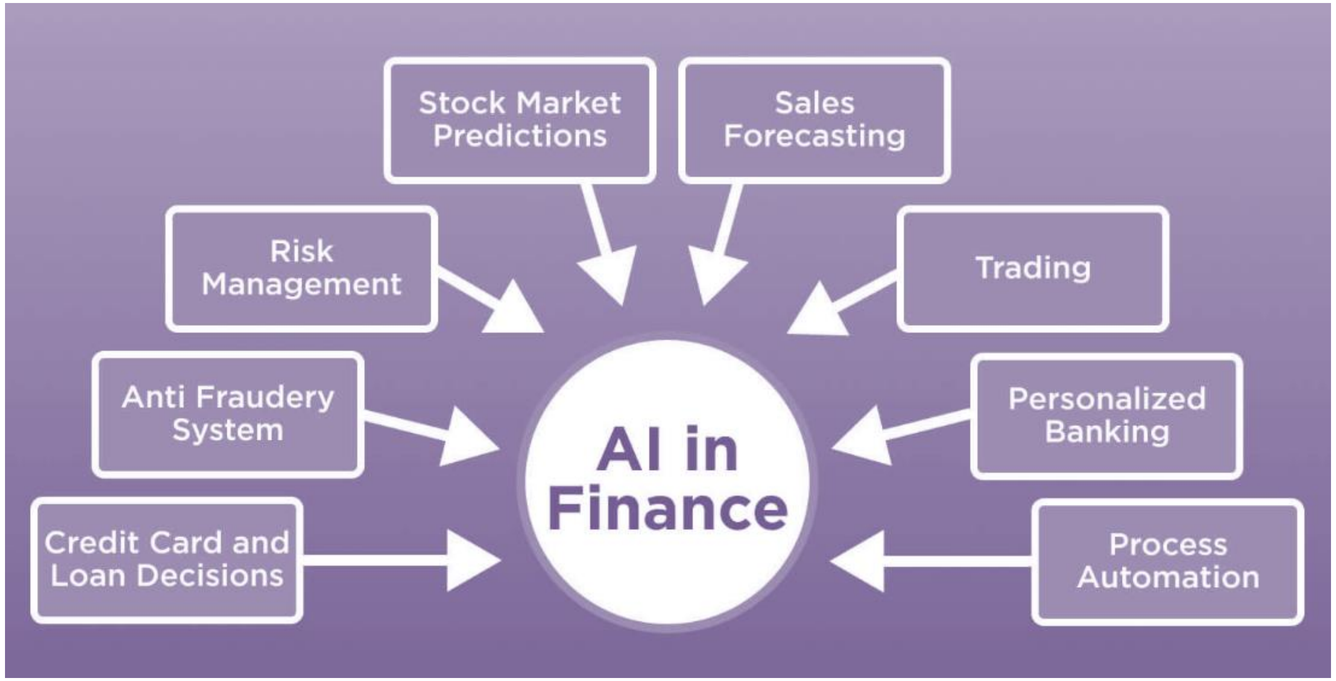


Usage of AI in Financial Services Spectrum



Top Use Cases of AI in Banking

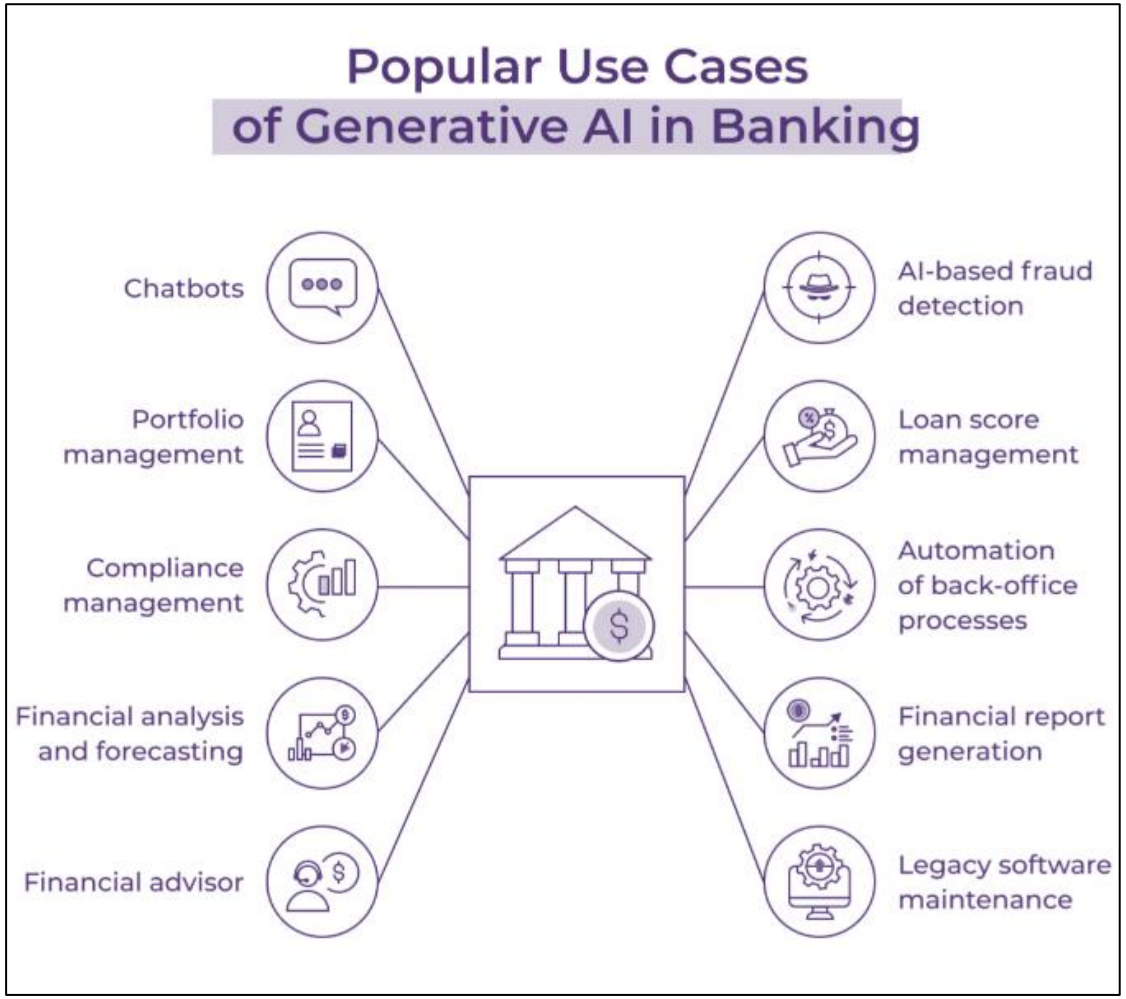
1. **Credit Scoring:** AI improves credit risk assessment using diverse data sources.
2. **Algorithmic Trading:** AI-driven algorithms optimize investment decisions in real-time.
3. **Risk Management:** AI predicts and mitigates financial risks efficiently.
4. **Customer Relationship Management:** AI enhances customer engagement and retention strategies.
5. **Fraud Detection:** AI algorithms identify and prevent fraudulent activities effectively.



Gen AI In Banking Industry

Gen AI can help in -

1.**Enhanced Customer Service:** AI-powered chatbots and virtual assistants provide personalized customer support, streamline inquiries, and offer 24/7 assistance, improving overall customer experience and satisfaction.
2.**Fraud Detection and Prevention:** AI algorithms analyze vast amounts of transaction data in real-time, identifying suspicious activities, detecting fraudulent patterns, and enhancing security measures to protect against cyber threats and financial fraud.
3.**Data-driven Decision Making:** AI-driven analytics tools leverage customer data to generate actionable insights, predict market trends, optimize investment strategies, and customize financial products, enabling banks to make informed decisions and improve operational efficiency.



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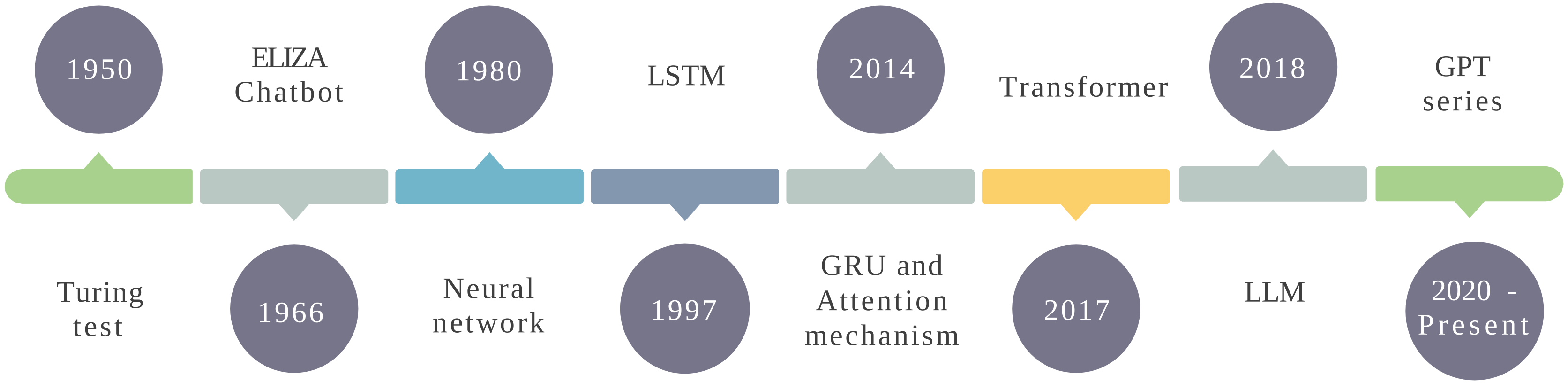
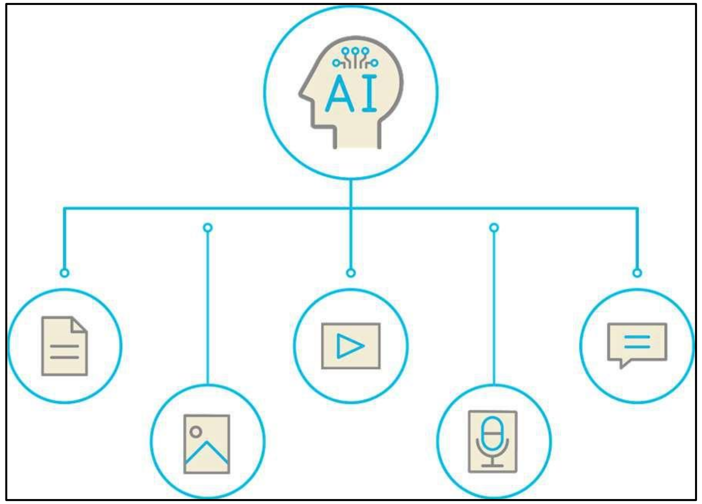
Evolution from AI to Generative AI

This timeline visually represents significant milestones in the field of artificial intelligence.



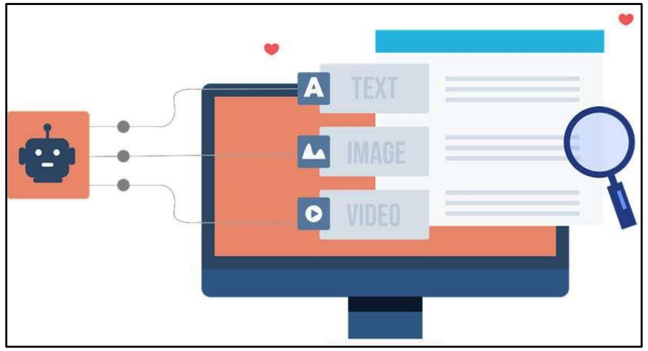
Generative AI



- Generative artificial intelligence (GenAI) is a category of AI systems that focus on generating new content, such as images, text, music, or other forms of creative output.

- It is a subset of artificial intelligence that leverages machine learning and deep learning techniques to generate data.

Objectives of Generative AI -



It is designed to independently generate new content by deriving its structure from recognized patterns.

It assists with complex problem-solving by generating numerous potential solutions, thus offering users a wide array of options.

It boosts human creativity by producing unique outcomes, enabling users to discover new ideas and possibilities.

Generative AI and LLM

- Large Language Models (LLMs) provide a foundation for generative AI by offering powerful, pretrained models capable of understanding and generating human-like text across various tasks and domains.
- Generative AI can serve as building blocks for more advanced models, enabling the development of innovative AI systems that create content ranging from natural language text to images and music.
- LLMs are state-of-the-art AI models designed to comprehend and generate human language and perform natural language processing tasks like Translation, Classification, and so on.

Large	Refers to the significant size and complexity of these models, which contain hundreds of millions or even billions of parameters
Language	Denotes their primary function, which is to understand and generate human language
Model	Describes them as mathematical representations that capture the patterns and structure of language data

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Widely Used LLMs

Model	Developer	Key features
GPT 3.5/4	OpenAI	It helps generate content, text summarization, sentiment analysis, data extraction, and language translations. The GPT 4 upgraded version bolsters multimodal capabilities, including the power to process different formats other than texts, like images and videos.
Gemini	Google	It aids in complex problem-solving and coding generation.
Llama	Meta AI	It enables natural language understanding, content creation, and language translation.
Falcon	Technology Innovation Institute (TII)	It provides language translation, common-sense reasoning, and sentiment analysis.
Claude	Anthropic	It supports open-ended dialogues, coding tasks, and data analysis of visual inputs like charts and graphs.

GenAI Tools

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Understanding Prompt Engineering

The Key to Effective AI Interactions

What Is a Prompt?

A prompt is a natural language text that instructs generative AI models to perform a specific task.

Here are some key reasons why prompts are essential:

- A precise prompt can lead to a more focused and useful output.
- A well-crafted prompt guides the GenAI to understand the context and the specific requirements of the task at hand.
- The prompt serves as a bridge between the user's intent and the AI's execution.

Important Components of a Prompt

Four important components that make up a great prompt are as follows:

- **Task** – Specific action or process
- **Instructions** – Rules to follow to achieve the desired outcome
- **Context** – Background information or situation that surrounds a task
- **Input** – Images, text, and dataset

Important Components of a Prompt: Example

Consider the below example to understand the important components of a prompt better:

ChatGPT

Create a brief narrative about a wizard's quest in a magical forest, adhering to the guidelines of incorporating a surprise ending, and draw from provided lore and creature descriptions.

Task:	**Create a brief narrative.**
Instructions:	**The story must involve a wizard's quest** and include a surprise ending.
Context:	The narrative is set in a **magical forest**, with the protagonist on a quest.
Input:	Use the **provided lore and descriptions of creatures** within the forest.

7 Types of Prompts

Explanation Prompts:

"Can you explain the concept of artificial intelligence in simple terms?"

"Provide a step-by-step guide on setting up a WordPress website."

Calculation Prompts:

"Calculate the area of a circle with a radius of 5 units."

"Determine the total cost of five items priced at $20 each."

Recommendation Prompts:

"Suggest a list of books for someone interested in learning machine learning."

"What are some top-rated restaurants in New York City for Italian cuisine?"

Comparison Prompts:

"Compare and contrast the advantages of Python and Java for web development."

"Explain the differences between a traditional classroom and online education."

Problem-Solving Prompts:

"Help me troubleshoot why my internet connection is slow."

"Guide me through resolving a '404 Not Found' error on a website."

Contextual Prompts:

"Given the context of a job interview, provide tips for answering common interview questions."

"Considering I'm planning a trip to Paris, suggest must-visit attractions and travel tips."

Interactive Prompts:

"Let's have a conversation about your favorite book. Start by summarizing the plot."

"Imagine you're a travel agent. Plan a two-week itinerary for a client exploring Southeast Asia."

5 Rules of creating effective prompts

1. Define a Persona
2. Define the Task
3. Set Constraints
4. Define the tone
5. Support with examples

4 Additional Phrases for effective prompts

“Let’s think step by step”

- "Let's think step by step about how a computer boots up. Start with the power-on sequence."
- "We're troubleshooting a network issue. Let's think step by step, beginning with checking physical connections."
- "Explaining the scientific method to a student. Let's think step by step, starting with observation and hypothesis."
- "To understand calculus, let's think step by step about the concept of limits and their importance."
- "When writing an essay, let's think step by step about the structure: introduction, body, and conclusion."
- "Solving a complex math problem. Let's think step by step, breaking down the problem into smaller parts."
- "In programming, let's think step by step about the process of debugging and finding code errors."

“Thinking Backwards”

The phrase "Thinking Backwards" can be used as a prompt to encourage creative problem-solving, critical thinking, or considering a situation from a different perspective.

Algorithm Design: "When designing an algorithm, start 'thinking backwards.' Begin with the desired output and work your way back to the input requirements.“

Product Design: "In product design, consider 'thinking backwards.' What user problem are we solving, and how can we design a solution that addresses it effectively?“

Programming: "In programming, the concept of 'thinking backwards' can involve starting with the desired output and then writing the code to achieve it."

“Act as a”,
“Pretend you are”,
“In Style of”

Act as a Financial Advisor: "Act as a financial advisor and provide investment advice for a client looking to diversify their portfolio."

Act as a Space Explorer: "Act as an astronaut and describe the experience of walking on the surface of the Moon during the Apollo 11 mission."

Act as a News Anchor: "Act as a news anchor and report on recent developments in renewable energy technology.

“Play Devil’s Advocate”

The prompt "Play Devil's Advocate" is often used to encourage someone to take a contrary or opposing viewpoint in a discussion or argument

Product Development: "Play Devil's Advocate and critique the proposed features of our new product from a customer's perspective. What might they not like about it?"

Legal Discussion: "Play Devil's Advocate and argue the case against a defendant even if you personally believe they are innocent."

Business Strategy: "Play Devil's Advocate and question the viability of our proposed market expansion plan, highlighting potential risks and drawbacks.“

Personal Growth: "Play Devil's Advocate with yourself. Challenge your own beliefs and decisions to gain a deeper understanding of your choices.

Market Research and Article Writing

12ft.io- Break Price Firewall for paid articles

libgen.si – Read e-books for free

Elicit.com – Search about Research Papers

Connectedpapers – Find out research papers

Sci-hub – Learn thesis for free

AI Tools for Presentation

- Slidegpt.com
- Sendsteps.ai
- Slidesgo.com
- Tome.app
- Powermode.ai

POWERFUL OPENING

TELL A STORY

LOW TEXT

USE VISUALS EFFECTIVELY

USE HUMOUR

KISS- KEEP IT STUPID SIMPLE

USE DATA & STATISTICS

General AI Tools

Formula Dog – To Create Excel Formula

Quillbot – To remove plagiarism and para phrasing

Askyourpdf + Merlin Extenstion

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Generative AI for Business

Session: 4

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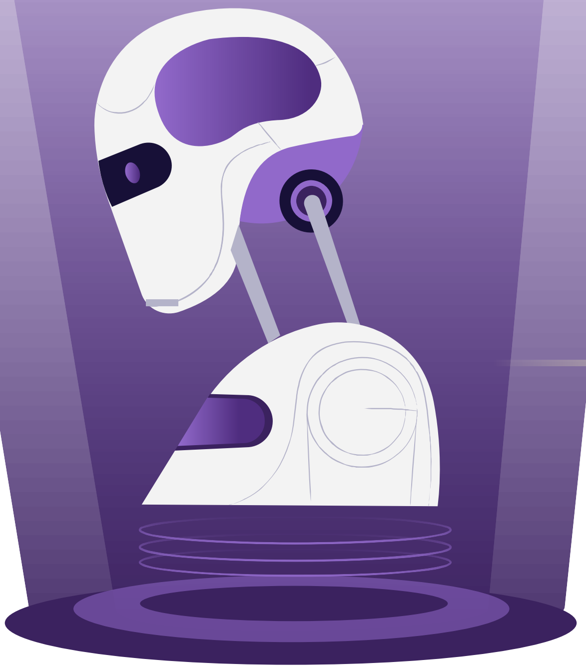
Workflow and Task Automation



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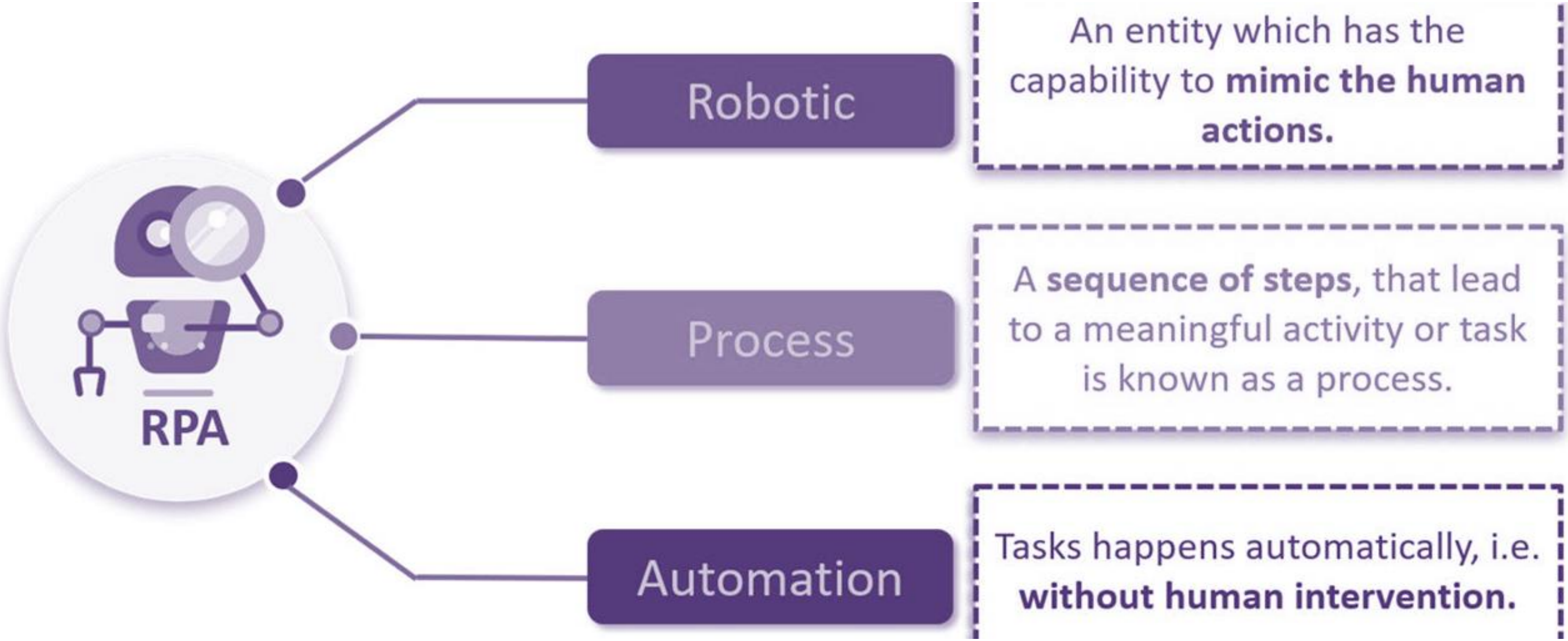
RPA

Understanding RPA (1/2)



Understanding RPA (2/2)

1. is the technology that allows computer software to mimic actions typically performed by humans.
2. is non-invasive and scalable
3. can work in any industry, especially when combined with AI.
4. reduces costs rapidly and increases ROI.
5. can automate highly manual, repetitive and rule-based processes, with low exception

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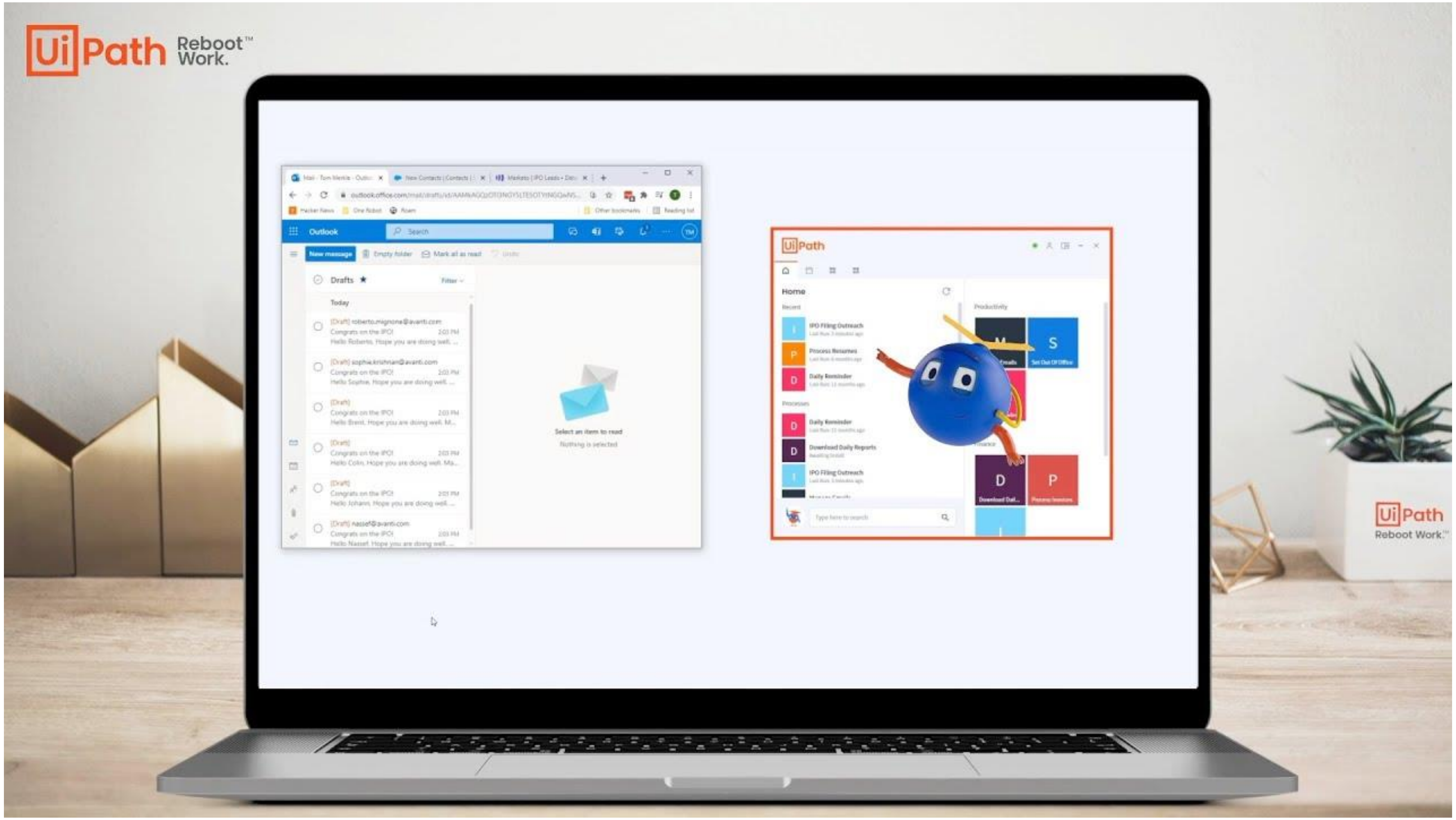
General Use Case

https://youtu.be/3R67RGjZoOM?si=lNmUefblb9P7QIUC



RPA – Facts & Figures – Gartner Report

1. By 2025, 80% of RPA-centric automation implementations will derive their value from complementary technologies.
2. By 2027,Automation Industry Size will become 12 Bn $, Currently at 6.8 Bn $
3. By 2026, there will be an 80% increase in the use of RPA for front-office functions to enable Universal Adoption in next 5 years
4. By 2024, organizations will lower operational costs by 30% by combining hyper automation technologies with redesigned operational processes

https://smartbridge.com/rpa-2020-findings-gartner-introduction-hyperautomation/

Manual work which can be automated

- Log in to any application
- Move files and folders
- Read and write to databases
- Scrape data from the web
- Connect to system APIs
- Extract content from documents, PDFs, emails and forms
- Open emails and attachments
- Make calculations

Daily Transactions	•Journal Entries •Reporting
Month End Close	•Open/Close Accounting Periods •Maintain Exchange Rates •Mid-Close Reconciliation •Financials and Calendar Close
Reporting	•Operational reporting •Enterprise reporting

Areas of Automation

01 **Email Automation**
Let the bots read and write emails seamlessly.

02 **Excel Automation**
Bots can read, write, merge and perform almost all actions using Excel.

03 **PDF Automation**
Bots can be used to read text, read with OCR, extract pdf range etc.

04 **Document Understanding**
Read the handwritten text via OCR engines.

05 **Manage Testing & Validation**
Manage validation and accelerate release cycle allowing you to automate testing.

06 **Integration Service**
Allows you to integrate with various other systems using library of connectors

Note: This are few of the main points. There are numerous other automation opportunities!

Top Players in the Automation Market

Figure 1: Magic Quadrant for Robotic Process Automation

Source: Gartner (August 2023)

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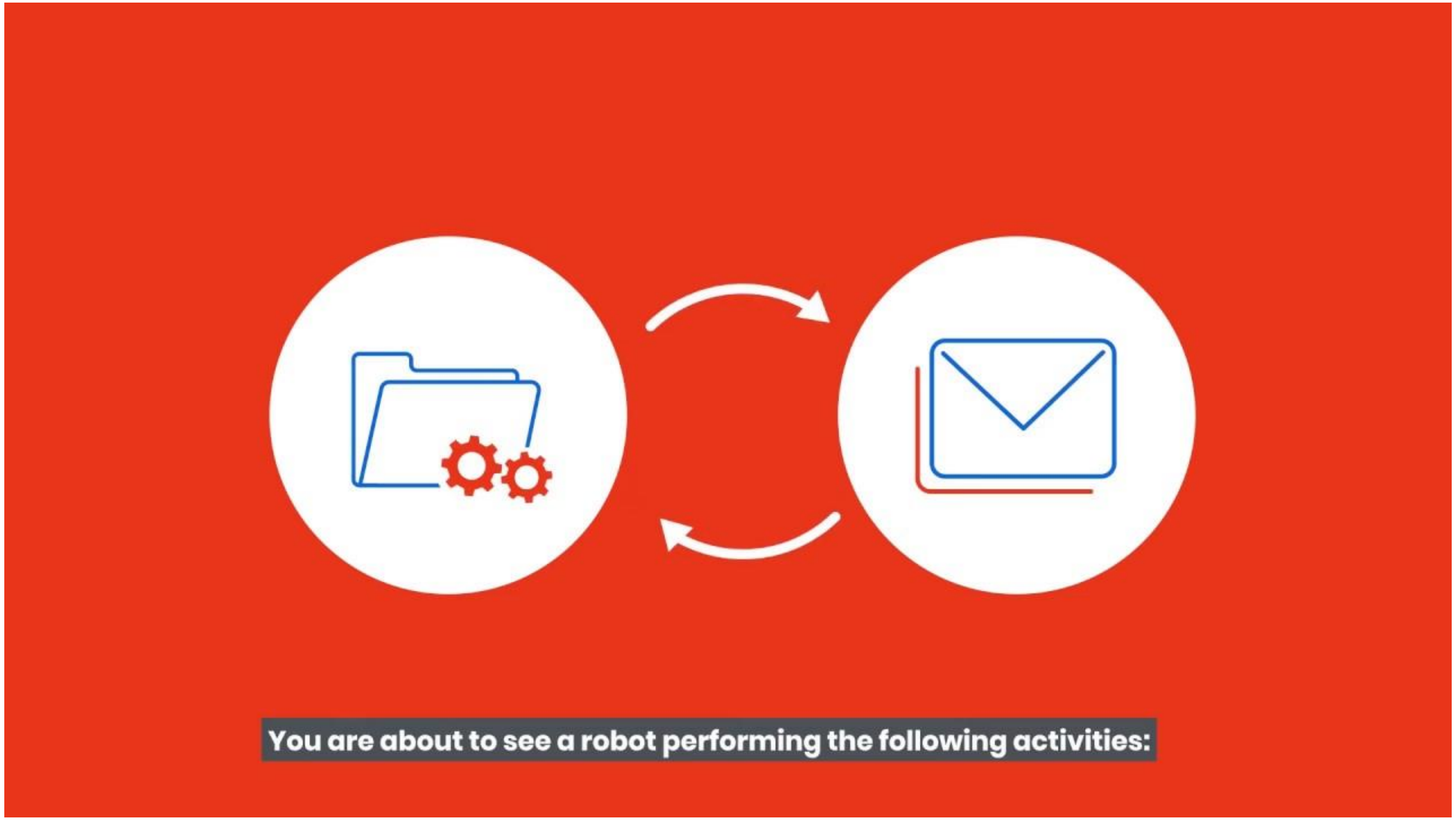
Industry Use Case - Invoice Processing Demo

https://youtu.be/10_MkXNMpVw?si=AnzIumPVS1whf2vh



Financial Accounting – Automation Use Cases

Record to Report	Procure to Pay	Order to Cash	Acquisition to Retire	Tax and Treasury	Strategic Reporting

1. General ledger reconciliation
2. Intercompany reconciliation
3. Financial Close activities
    1. Open Close Accounting Periods
    2. Maintain monthly exchange rates
4. Manual journal entries
5. Bank reconciliations
6. Intercompany settlements
	1. GR/IR Analysis and Reconciliation
	2. PR to PO Conversion
	3. Vendor invoice processing
	4. 2-way/3-way match reconciliation
	5. Vendor notifications
	6. Accrual journal entries and reconciliation
		1. Sales order creation
		2. Sales order updates
		3. Customer invoices generation
		4. Customer notifications
		5. Monthly reconciliation (unbilled, deferred)
		6. Closing activities
			1. Fixed Assets Creation
			2. Mass updates
			3. Monthly closing runs
			4. Periodic reconciliation
				1. Data aggregation for filings
				2. Bank reconciliation
				3. Prepare tax returns
				4. Monthly reconciliation
					1. Financial statements
					2. Aggregation of data
					3. Management reports
					4. Excel processing

Top Use Cases in Banking Industry

Transformation Segments

HUMAN

Transform relationships

Using IA, people will be able to spend more time on exceptional work: the 20% of non-routine tasks that drive 80% of value creation.

PROCESS

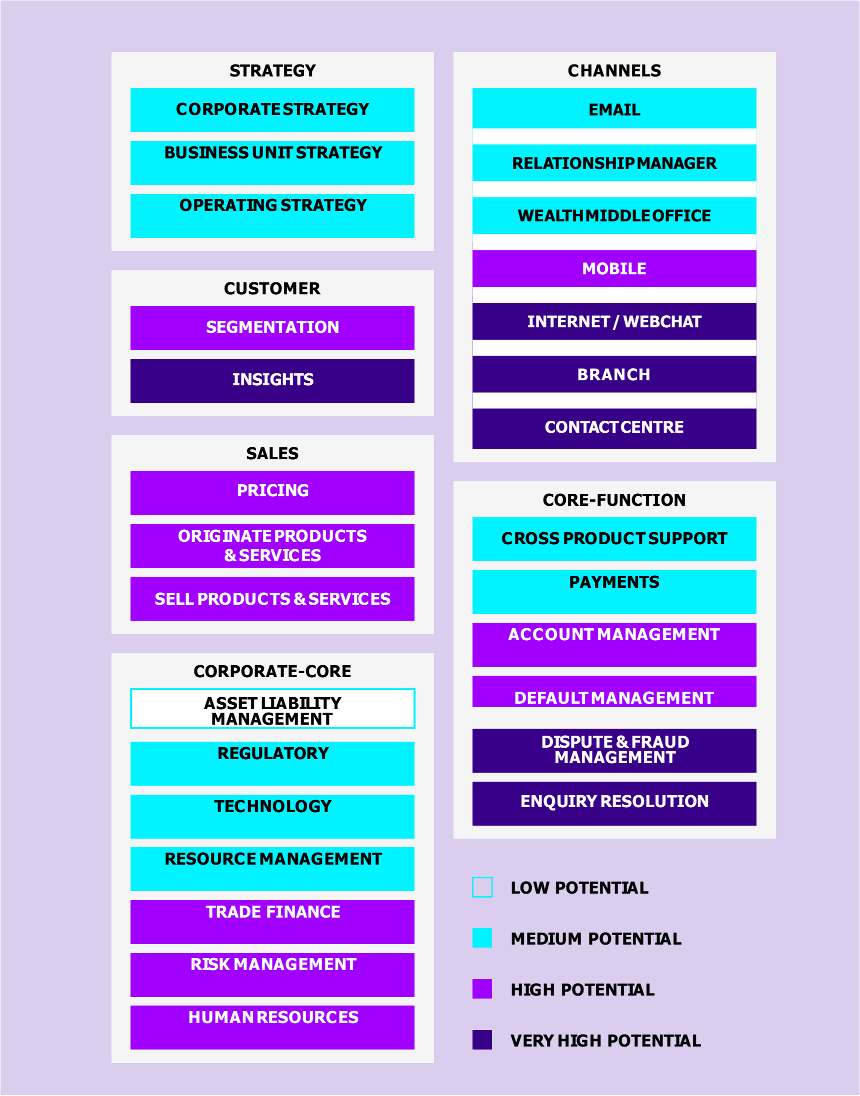
Re-imagine business models and processes

Smart machines will continually review end-to-end processes and apply 'intelligent automation of process change' to refine and optimize.

DATA

Illuminate dark data

Companies will apply IA to greatly enhance large data analytics, evolve algorithms with transactional data faster, and combine data in new ways to discover trends.



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Examples (1/3)

Process – STO (Stock Transport order)

THE CHALLENGE

- STO requires effective communication between multiple parties & location.
- Delay in STO may lead to missed delivery, stock outs & excess inventory.
- Tracking and traceability in supply chain is complex and essential.

1

THE SOLUTION

- Unattended BOT reduced manual process by 98% and 20 FTE benefit realized across multiple departments
- RPA BOT created by us automated various below aspects of STO process
  - Order Creation
  - Data validation - reconciliation
  - Inventory management

Process – EOD Settlement and Tracking (Finance Process)

THE CHALLENGE

- The avg. settlement time was 3-4 hours.
- Tracking of the Settlement and monitoring the settlement amount was a tedious process

2

THE SOLUTION

- Smooth integration between multiple application like SAP, Excel, Macros, CRM tools and network drives.
- Unattended robot BOT was configured, and the manual processes were reduced drastically.

Examples (2/3)

Process – SAP Data Extraction, Reporting and Reconciliation

3

THE CHALLENGE

- Manual data extraction – running SAP reports (for example GL Line-item Report)
- Download in excel and reconciliation which is time consuming and inefficient
- User dependency for time sensitive tasks

THE SOLUTION

- Unattended robot running at scheduled interval to download SAP data (direct table extract or through GUI), excel processing and building monthly file
- Sending notification to end user with variance
- Efficient and on-time closing

Process – Open and Close Accounting Periods

4

THE CHALLENGE

- User dependency (to open calendars first of the month, regardless of weekend, holiday)
- No value-added task including mass notification

THE SOLUTION

- Unattended robot running 1st of the month (as scheduled, 1am) to change FI, CO and MM periods and send notification
- Unattended bot running last day of the close (using slack bot) to close FI, CO and MM period and send notification

Examples (3/3)

Process – Purchase Requisition Status (for requisitioners)

THE CHALLENGE

- End user (requisitioner) not getting direct notification about PR status (Approved, PO issued, Invoiced or not etc.)
- Required them to login to SAP, navigate screens to get the information
- End user training and learning curve

5

THE SOLUTION

- Unattended robot triggered via Slack
- Bot in the backend establish connectivity with SAP as API calls, get PR status, formats it in excel and send email attachment to requester
- Bot also checks any pending GR or IR and accordingly notify AP clerk or PO requester

Process – Sales order updates (Bill Plan)

THE CHALLENGE

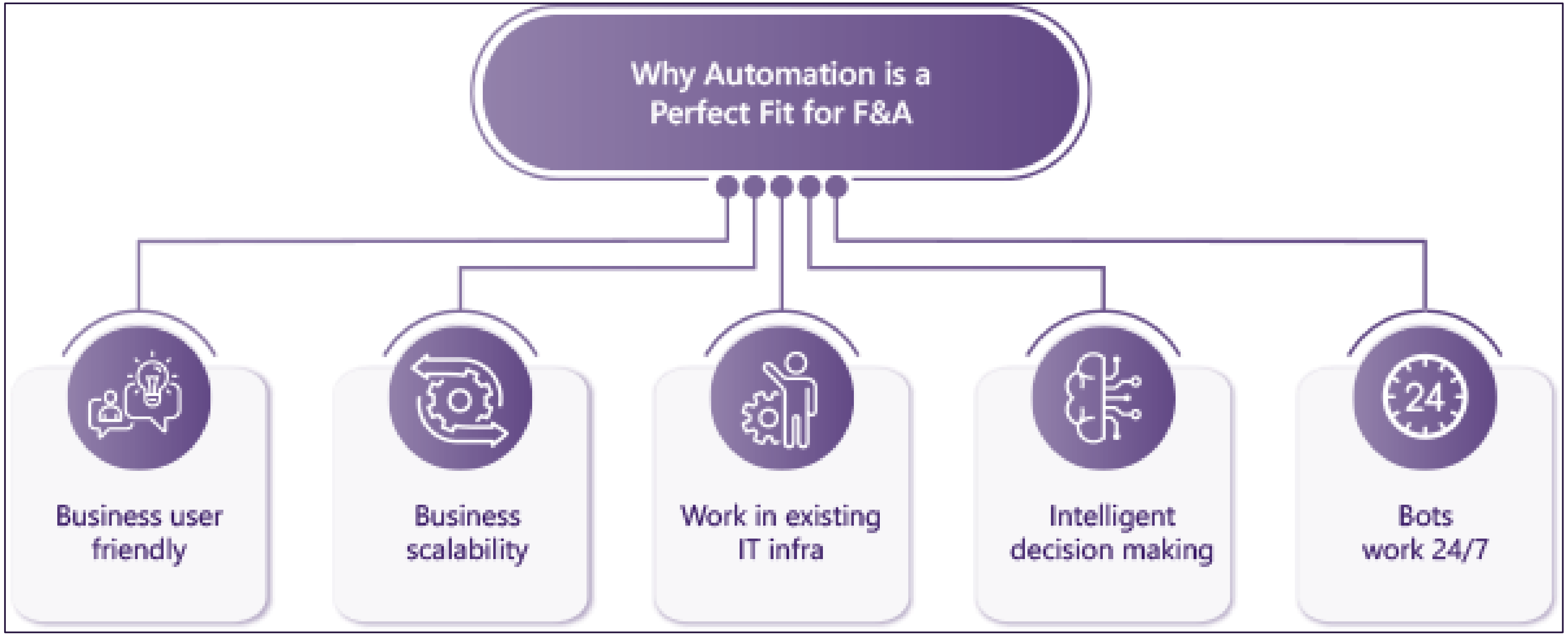
- Manual exercise to update sales order line items (for multiple vendors)
- Time consuming

6

THE SOLUTION

- Attended bot for excel data processing to trigger VA02
- Bot process through SAP Screens to make necessary updates
- Bot update excel file back with status for reconciliation
- Bot notify user for any errors and keep track of pending transactions

Benefits of RPA



Pedagogy Approach

PHASE 1	PHASE 2	PHASE 3
Theory	Practical	Live
Introduction Concepts Detailed Understanding Complete Overview	Hands-On Case Studies Lab Exercise Business Case Building DIY	Industry Connects Mentorship Live Projects Internships

How to learn Automation and Profiles Pathway

RPA Overview

GENERAL
(For Both Tech & Non-Tech Students**)**

RPA Developer

SPECIFIC to Technology Students

RPA Business Analyst

SPECIFIC to Non-Tech Students

UiPath StudioX Installation - https://youtu.be/W-cFCwQRqao?si=RgD-FL-dzbT13zI7
Citizen Development on academy.uipath.com - https://academy.uipath.com/learning-plans/rpa-citizen-developer-foundation
Bot Demo Link - https://www.youtube.com/watch?v=JzhKh3nAduQ&t=389s

Thank you for your attention

Any questions?